Data Engineering
ASSIGNMENT - 05

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1. Techniques for Modeling Batch Analytical Data - Inmon, Kimball, and Data Vault

Ans Batch analytical data can be modeled using Inmon, Kimball and Data Vault techniques, depending on business needs

* Inmon Model

- Top-down approach
- Creates a centralized Enterprise Data Warehouse (EDW)
- Uses highly normalized tables (3NF)
- Data marts are created later.

Advantages
- Strong governance
- Single source of truth
- Good data consistency

Limitation
- Slow development
- Complex for reporting

* Kimball Model

- Bottom-up approach
- Uses dimensional modeling
- Built with Fact and Dimension Tables
- Mainly uses Star Schema.

Advantage
- Fast query performance
- Easy BI and reporting

Disadvantage
- Data redundancy
- harder enterprise-wide integration.

* Data Vault Model

Uses:
- Hubs → business keys
- Links → relationships
- Satellites → description history

- Faster dashboard queries
- Easy for analysts
- Better performance for OLAP workloads.

* Materialized Views

stores the result of a query physically. Instead of calculating joins and aggregates every time, results are precomputed.

Benefits

- Very fast repeated queries
- Good for summaries and KPIs

Limitation

- Requires periodic refresh
- Slight data latency.

* Federated Query

allows querying external data sources directly from a warehouse without moving data.

Benefits

- No extra ETL
- Useful for hybrid cloud data

Limitations

- Depends on source system speed
- Network latency may slow query.

* Query Virtualization

creates a logical layer over multiple sources without storing data internally. It often uses query pushdown to send computation to source system.

Advantages

- high scalability
- excellent audit trail
- Easy schema evolution

Limitations

- Complex joins
- Not easy for direct analytics.

6 (a) Business Analytics and Operational Analytics

Business analytics focuses on historical data, trends, KPIs and strategic decision-making

examples :-

- Sales reports
- Customer churn analysis
- Forecasting & Dashboard reporting

Advantages

- helps management decision
- Identifies long-term trends
- Improves business strategy.

Operational Analytics focuses on real-time or near-real-time actions

examples

- Fraud detection
- Live inventory monitoring
- System alerts
- Delivery tracking

Advantages

- Fast-decision making
- Immediate problem detection
- Supports real-time systems.

b) Methods and Systems used to Serve Data for Analytics and ML

An Methods

1. File Exchange - CSV, Parquet, JSON files
2. Databases / Data Warehouses - structured storage for BI
3. Streaming Systems - real time pipelines
4. Query Frederation - Querying multiple source together.
5. Semantic and Metrics Layers - reusable business definitions.
6. Notebook Serving - Jupyter notebooks for analysis
7. Feature Stores - ML - ready features
8. Reverse ETL - send processed data back to business tools.

Systems

- Snowflake
- Google BigQuery
- Amazon Redshift
- Apache Kafka

4. Describe in detail the different techniques used in improving the performance of a query with emphasis on prejoining data and Materialized Views, Federated and Query Virualization.

An PreJoining Data

means joining frequently used tables in advance and storing them as denormalized table.

Benefits

- Reduces repeated join cost.

Benefits

- Unified acasse layer
- Better flexibility
- less data duplication.

Limitation

- Slower than local tables
- Performance depends on source.